

PAPER ID	:	#17 (1571278052)
PAPER TITLE	:	Enhancing Pedagogical Feedback Quality of Large Language Models via Learning Analytics in a Kanban-Based Platform for Student Self-Regulated Learning Support

Reviewer Feedback

Go through each comment one by one — a page or paragraph reference makes it easy for everyone to follow.

NO.	REVIEWER'S FEEDBACK	AUTHOR'S RESPONSE	LOCATION OF REVISION IN MANUSCRIPT
Reviewer 1			
1	On dataset size and generalizability <i>"The evaluation is based on a relatively small synthetic dataset (40 instances), which may limit the generalizability of the results."</i>	We agree, and made this constraint explicit and auditable rather than implicit. Section III-C now documents the full dataset generation procedure (five behavioural profiles and their counts, the rules governing card counts and checklist-completion percentages, domain coverage, prompting techniques, and generator model). The limitation is now stated plainly in the Discussion and Conclusion, and Future Work commits to replicating the experiments on real student Kanban interaction logs.	Section III-C (Dataset Construction); Section IV-D (Discussion); Section V (Conclusion)
2	On the absence of statistical significance testing <i>"Furthermore, the absence of statistical significance testing makes it difficult to assess the robustness of the reported improvements."</i>	We accept this. Within the six-page camera-ready limit we could not add a full statistical-testing section, so instead we calibrated the paper's claims to the strength of evidence available: the narrow BERTScore/BARTScore margins are now explicitly labeled exploratory, and only the larger LLM-as-a-Judge and expert-rated	Section IV-D (Discussion); Section V (Conclusion, Future Work)



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		differences are interpreted with confidence. Revised text (Future Work): <i>"apply paired significance testing over per-instance scores (paired t-test or Wilcoxon signed-rank with bootstrap confidence intervals) to establish which margins are reliable"</i>	
3	On the scale of expert validation and inter-rater agreement <i>"The expert validation is also limited in scale, and reporting inter-rater agreement would strengthen the credibility of the evaluation."</i>	We agree; with a single validator, inter-rater agreement cannot be computed at all, so expanding the panel is the necessary first step. This limitation is now stated explicitly in the Discussion and Conclusion, with the remedy committed to in Future Work.	Section IV-D (Discussion); Section V (Conclusion, Future Work)
4	On deeper interpretation of model behaviour and linkage to prior literature <i>"The discussion could be further improved by providing deeper interpretations of the observed model behaviors... Relating the findings more explicitly to prior literature and exploring potential explanations for these performance differences would enhance the scientific contribution of the paper."</i>	This prompted the most substantial revision. The single unsupported conjecture explaining the 70B model's degradation under ReAct+LA has been replaced with two literature-grounded explanations: (1) an over-prompting effect, where added scaffolding degrades performance once a model already possesses the required competence [18]; (2) the implicit-Bayesian-inference view of in-context learning [16], under which a model with an already-concentrated prior gains little from added prescriptive context. The second explanation also ties back to the paper's own theoretical framework.	Section IV-D (Discussion), "Model size" paragraph
5	On practical significance and the implications of the limitations <i>"...a more detailed discussion of the practical significance of the improvements and the implications of the study's"</i>	A new paragraph was added to the Discussion addressing both halves of this comment: it frames the +0.075 F1 gain as the practically meaningful result (recovering part of	Section IV-D (Discussion), new paragraph beginning "Practically, the gains matter most..."



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	<i>limitations would strengthen the overall discussion."</i>	the performance gap to the untuned 70B model at a fraction of the deployment cost), and states plainly that the synthetic dataset and single validator mean the results establish feasibility and direction rather than deployment-ready effect sizes.	
Reviewer 2			
1	References and DOI <i>"The author is encouraged to include a DOI for every reference to ensure traceability and ease of verification for each source."</i>	Implemented as far as the sources permit. Several references already had DOIs in our bibliography but were not rendering in the compiled PDF due to the bibliography style; this is now fixed. Missing DOIs were then looked up and added (arXiv, ACL Anthology, Zenodo, and others). In the revised manuscript, 24 of the 25 cited references now display a DOI; the single remaining source (Groq API Pricing) is a vendor webpage with no registered DOI, so its URL is retained instead.	Reference list (all 25 entries), final page
2	Fig. 1, diagram clarity <i>"The direction of the arrow between 'Feedback Agent' and the C1 to C5 block is unclear... Please add an explicit arrowhead and/or a short label... to remain consistent with the flowchart notation already used elsewhere in the same figure."</i>	Implemented. Figure 1 has been redrawn so the connector between the Feedback Agent and the C1-C5 block carries an explicit arrowhead with a labeled direction, making it unambiguous that C1-C5 are components composed into the agent's prompt (not outputs), consistent with the notation used elsewhere in the figure.	Fig. 1 and its caption, Section III-D
3	Statistical significance testing <i>"Without a statistical test... there is no basis to determine whether these differences are statistically meaningful or fall within the range of random</i>	We accept this. As with Reviewer 1's related comment, we could not add a full testing section within the page limit, so we reduced the claims to match the evidence: the	Section IV-D (Discussion); Section V (Conclusion, Future Work)



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	<i>variation..."</i>	central finding no longer rests on the small BERTScore margin but on the LLM-as-a-Judge gain (+0.075 F1, an order of magnitude larger), which is also corroborated by the independent expert validation. Paired significance testing is now a named, committed next step in Future Work.	
Reviewer 3			
1	On the monotonic improvement claim <i>"The claim that the 8B model improves monotonically with increasing context complexity should be revised, because the Material List condition performs slightly worse than the Baseline on both BERTScore and BARTScore."</i>	The reviewer is correct; the unqualified claim was inconsistent with our own Table 1 and has been corrected throughout. The Results section now explicitly reports the dip at Material List for BERTScore/BARTScore. The monotonicity claim is retained only where it genuinely holds: the 8B model's LLM-as-a-Judge F1 (0.5371, 0.5627, 0.5896, 0.6121), which increases at every step. Abstract and Conclusion wording were revised to scope the claim to this metric explicitly.	Abstract; Section IV-A (Results); Section V (Conclusion)
2	On details of dataset generation <i>"The dataset is relatively small and fully synthetic. The authors should provide more details about how the 40 synthetic Kanban datasets and the five behavioral profiles were generated."</i>	Added. Section III-C now documents the generation procedure in full: the five behavioral profiles and counts, the OOP domain and subtopics, the role-play/few-shot/Chain-of-Thought prompting techniques used, the generator model (ChatGPT-5 Thinking), and the provenance of the ground truth (three university lecturers in educational psychology).	Section III-C (Dataset Construction and Model Selection)
3	On the scope of expert validation <i>"The expert validation is limited to one independent expert and</i>	We agree; this converges with a related comment from Reviewer 1. Recruiting a second independent	Section V (Conclusion, Limitations and Future Work)



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	<i>only one selected configuration. It would be stronger to involve multiple experts and compare more than one model configuration."</i>	expert and running a second validation round was not achievable within the revision window, so the constraint is now stated plainly as a limitation rather than understated, with the remedy committed to in Future Work.	
4	On the selection of the "best-performing configuration" <i>"The reason for selecting Llama 3.1 8B + ReAct + LA as the 'best-performing configuration' should be clarified, because the 70B Material List configuration achieves higher BERTScore and LLM-as-a-Judge scores."</i>	The reviewer is right that the original label was misleading. The terminology was corrected throughout: the 8B+ReAct+LA configuration is now consistently described as the top-performing 8B configuration, never as the best configuration overall, and the Results section states plainly that the 70B Material List variant achieves the best BERTScore and LLM-as-a-Judge scores across both models. The selection rationale for qualitative review is now stated explicitly at the point of use.	Abstract; Section I (Introduction); Section IV-C (Qualitative Expert Validation); Section V (Conclusion)
5	On the strength of the model-size conclusion <i>"The conclusion that model size fundamentally determines optimal context complexity is too strong. Since only two different Llama versions are compared, the observed effect cannot be attributed solely to parameter size."</i>	We accept this and weakened the claim to match the evidence: "fundamentally determines" has been replaced with "strongly shapes" everywhere it appeared. We also now name the confound directly: the two systems differ in release version as well as parameter count, so the effect cannot be attributed to scale alone; this is stated as a limitation ("two model scales within one family"), with disentangling version from scale listed as future work.	Section I (Introduction, Contributions); Section V (Conclusion, Limitations)



Desk Reviewer Feedback

Any editorial notes from the desk review team.

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1	No desk reviewer comments were provided for this paper.	—	—

All revisions were made within the six-page IEEE camera-ready limit. We thank all three reviewers and the desk review team for their constructive and detailed feedback, which has substantially strengthened the manuscript.

